# Semantic Information Extraction of Energy Requirements from Contract Specifications: Dealing with Complex Extraction Tasks

Peng Zhou[1] and Nora El-Gohary[2]

**Abstract:** Automated specification energy compliance checking aims to check the compliance of building designs captured in building information models (BIMs) with energy requirements from contract specifications. However, automated extraction of requirements from specifications to support automated compliance checking is challenging because of the text complexities of the specifications, including hierarchically complex text structures, incomplete sentence structures, and a variety of levels of development (LODs). To address this challenge, this paper proposes a semantic information extraction method to extract building energy requirements automatically from specifications. Specifically, three new submethods are proposed to deal with these complexities: a domain-specific text splitting and stitching method, an incompleteness-aware sequential dependency extraction method, and a detail-aware LOD extraction method. The proposed information extraction method was tested in extracting thermal-insulation and lighting-power requirements from MasterFormat specifications. A performance of 96.8% recall and 97.6% precision was achieved on the testing data, which indicates the effectiveness of the proposed method. **DOI: 10.1061/(ASCE)CP.1943-5487.0001008.** © *2022 American Society of Civil Engineers.*

**Author keywords:** Information extraction; Ontology; Automated compliance checking; Contract specifications.

## Introduction

Specification energy compliance checking aims to help construction projects comply with energy requirements in contract specifications for achieving building energy efficiency. Contract specifications prescribe owner requirements that are in addition to the regulatory requirements in the codes. Therefore, specification checking is essential for meeting the contractual obligations for the project. However, manual compliance checking is time-consuming. Thus there is a need for automated approaches to check the compliance of building designs, represented in building information models (BIMs), with energy requirements in specifications (Amor and Dimyadi 2021).

The most difficult component in automated compliance checking (ACC) is automated requirement extraction: automatically extracting the requirements from the text and representing these requirements in the form of computer-processable rules. A significant number of efforts have been undertaken in the area of ACC. These efforts have achieved different levels of automation in requirement extraction: little or minimal, semi, or full automation. Many efforts have reached little or minimal automation. For example, a large number of research efforts (e.g., Sydora and Stroulia 2020; Nawari 2020; Kim et al. 2019; Zhou et al. 2018; Solihin and Eastman 2016; Dimyadi et al. 2016; Preidel and Borrmann 2015; Cheng and Das 2014; Choi et al. 2014; Qi et al. 2014; Zhong et al. 2012; Nawari 2012; Pauwels et al. 2011; Tan et al. 2010) and software tools [e.g., CORENET (Khemlani 2005), Solibri Model Checker (SMC 2009), SMARTreview (2021), and ComCheck/ResCheck (USDOE 2018)] require manual extraction of requirements from text, which is time-consuming and unscalable. Few efforts achieved a level of semiautomation. For example, Häußler et al. (2021), İlal and Günaydın (2017), and Beach et al. (2015) semiautomatically extracted requirements from text that is annotated with predefined tags. However, the annotation itself requires substantial manual effort. Fully automated efforts, on the other hand, are very limited. For example, Zhang and El-Gohary (2013, 2017) and Zhou and El-Gohary (2017) proposed semantic information extraction (IE) methods to extract requirements automatically from the International Building Code and the International Energy Conservation Code, respectively. However, despite their contributions, their methods cannot address the text complexities in specifications.

Compared with IE from building and energy codes, automated IE from specifications faces three new challenging text complexities: hierarchically complex text structures, incomplete sentence structures, and a variety of levels of development (LODs). First, text in specifications usually is organized in a more complex—wider and deeper—text hierarchy. For example, a section in the specifications may contain many articles, in which each article contains many paragraphs, each paragraph contains multiple levels of subparagraphs, and each subparagraph contains multiple requirements. Second, requirements in the specifications usually are represented in short phrases, rather than complete sentences. Such incomplete sentence structures hinder the ability to capture the dependency between the semantic information in the text. Such dependency information helps reduce text ambiguities and enhance the extraction performance. Third, requirements in specifications may correspond to different BIM LODs. Extraction of requirements that go beyond the required LOD may result in unnecessary processing efforts and potential compliance checking errors.

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Note. This manuscript was submitted on June 24, 2021; approved on October 27, 2021; published online on July 6, 2022. Discussion period open until December 6, 2022; separate discussions must be submitted for individual papers. This paper is part of the ***Journal of Computing in Civil Engineering***, © ASCE, ISSN 0887-3801.

To address these text complexities, this paper proposes a semantic IE method to extract building energy requirements automatically from contract specifications. The proposed method is novel in four main ways. First, it addresses a less explored area in the literature, which is automated deep IE from specifications. In contrast to shallow extraction, deep extraction can extract all information that describes a regulatory requirement. Second, a domain-specific text splitting and stitching method is proposed to simplify the hierarchically complex text structures automatically using a regular expressions-based pattern-matching technique. Third, an incompleteness-aware sequential dependency extraction method is proposed to capture dependency information from incomplete sentence structures to reduce text ambiguities. Fourth, a detail-aware LOD extraction method is proposed to automatically differentiate the LODs of sentences based on analyzing their grammatical moods using syntactic text features.

## Background

### Text Characteristics of Contract Specifications

Different types of documents are written in different styles to convey information for different purposes. Contract specifications are relatively well-organized—following standardized text organization and formatting—and are written in a highly concise language to reduce verbiage. The MasterFormat is a hierarchical classification system developed by the Construction Specifications Institute (CSI) and Construction Specifications Canada (CSC) for organizing project manuals (including contract specifications) (CSI and CSC 2014). Specifications are divided into divisions, each division is broken down further into a number of sections, and each section consists of three parts: general, products, and execution (CSI and CSC 2009, 2017). The text in each part is organized in a hierarchical structure, in which each part consists of at least one article, each article contains at least one paragraph, and each paragraph may contain zero, one, or multiple levels of subparagraphs. An illustrative example of an article is shown in Fig. 1.

Sentence structures in contract specifications are different from those in energy codes in two ways. First, shorter and incomplete sentences are used in the specifications. A sentence usually is composed of one or multiple phrases. A phrase is a piece of text that is separated by a delimiter such as a comma, colon, or semicolon (such delimiters are the text features of incomplete sentence structures, which are called incompleteness features hereafter). Second, two grammatical moods—imperative and indicative—are used to write sentences in specifications, whereas only the indicative mood is used in codes.

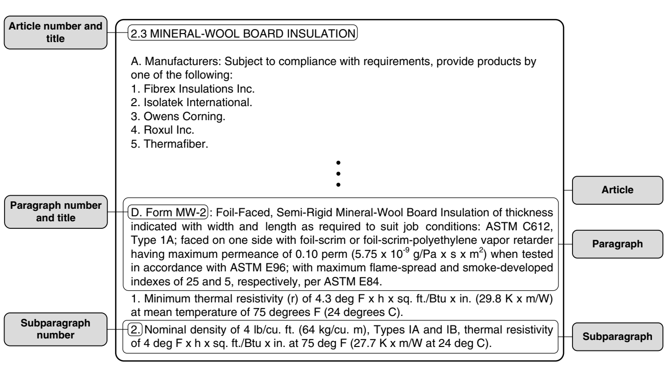


**Fig. 1.** Example article in specifications.

Imperative sentences are featured by a verb that explicitly defines an action (called action verbs hereafter) as the beginning of a sentence, in which the subject of an imperative sentence (e.g., contractor) is implicit to improve conciseness and understandability. Imperative sentences often are used to prescribe requirements related to the installation of products or equipment (CSI 2004; Kalin et al. 2010), and these requirements correspond to a BIM LOD of 400 or above. An imperative sentence that contains a fabrication requirement, which is in LOD 400 or above, is "Fabricate corners minimum 0.45 m × 0.45 m (18 in. × 18 in.) mitered and sealed as one piece." Indicative sentences are featured by a modal verb (e.g., shall) that precedes the main verb of a sentence. The use of indicative sentences is minimized because they may result in unnecessary verbiage.

Contract specifications use a streamlined writing style, which is a writing technique recommended by CSI to reduce verbiage and grammatical issues in writing specifications (CSI 2004; Kalin et al. 2010). Using streamlined writing, paragraphs (including subparagraphs) are written in a three-tuple format <topic, colon, content> in which the topic usually is a keyword or phrase that refers to the primary subject of a paragraph, the colon means shall, and the content contains a number of sentences that prescribe detailed requirements related to the topic. Streamlined-written paragraphs have unique text capitalization features (e.g., the first letter of each word in the topic and the first letter of the first word in the content are capitalized). Such capitalization features may be used to define text feature patterns.

### Information Extraction Approaches

There are two common IE approaches: machine learning-based and rule-based approaches (Jurafsky and Martin 2020; Moens 2006; Moreno et al. 2013). Supervised and semisupervised machine learning-based approaches require human developers to annotate manually the text features and corresponding target information for a large set of training data, and applies machine-learning algorithms to learn the extraction rules automatically from the annotated training data. In comparison, rule-based approaches require human developers to define manually the patterns of the text features from a small set of representative data (called development data hereafter) and develop a set of extraction rules based on the defined patterns. The features are syntactic [e.g., part-of-speech (POS) tags] and/or semantic features (e.g., ontology concepts), and the patterns commonly are defined using pattern-matching languages (e.g., regular expressions) (Jurafsky and Martin 2020). In this research, a rule-based approach was selected because of its performance. There is a need for an exceptionally high level of extraction performance to ensure sufficiently high performance in noncompliance detection—about 85% recall or higher, with good precision. Rule-based IE tends to achieve higher performance levels, especially for deep IE, because the human expertise usually results in more-accurate feature patterns (Moens 2006; Lee et al. 2019). Zhou and El-Gohary (2017) showed that rule-based IE is able to achieve a very high level of performance—over 95% recall and precision.

### Semantic Information Extraction in Construction Domain

Information extraction efforts are limited in the construction domain. Example efforts using a rule-based approach include those of Abuzir and Abuzir (2002), who used lexicosyntactic features and document structure features [i.e., HyperText Markup Language (HTML) tags] to extract terms and their relations from web pages for constructing a thesaurus of civil engineering; Al Qady and Kandil (2010), who used syntactic features (e.g., phrase segments) to extract concepts and relations from contract documents for enhanced document management; and Xu et al. (2021), who used lexical and syntactic features to extract domain knowledge elements (i.e., noun or relation phrases) from safety accident reports for construction safety management. Example efforts using a machine learning-based approach include those of Liu and El-Gohary (2017), who extracted information entities that describe bridge deficiencies and maintenance actions from bridge inspection reports for improved bridge deterioration prediction; Zhong et al. (2020), who extracted qualitative construction procedural constraints from construction quality codes; Zhang and El-Gohary (2021), who used transfer learning strategies for IE from multiple regulatory documents; and Xue and Zhang (2021), who proposed an enhanced POS tagger for improved IE.

A few efforts focused on IE from contract documents to support various applications. For example, Lee et al. (2019) used a rule-based approach to extract risk clauses in terms of subject–verb–object tuples from Fédération Internationale Des Ingénieurs-Conseils (FIDIC) specifications for supporting international contract management. Akanbi and Zhang (2020) used a rule-based approach to extract design information in terms of six named entities (e.g., material names) from MasterFormat specifications for supporting detailed wood construction cost estimation. Moon et al. (2021) used a machine-learning approach to extract five named entities (e.g., organizations) from Qatar road construction specifications for supporting construction specification review. Despite their contributions, the aforementioned efforts focused only on shallow IE, which refers to the extraction of partial information from a sentence (Zhang and El-Gohary 2013; Zhou and El-Gohary 2017). In contrast, deep IE, which aims to extract all information expressed by a sentence based on a full analysis of the sentence (Zhang and El-Gohary 2013; Zhou and El-Gohary 2017), is required for ACC. Other efforts dealing with contract documents (e.g., Hassan and Le 2020; Jeon et al. 2021) focused on text classification (TC), not IE.

Two main efforts used rule-based approaches for deep IE to support ACC. First, Zhang and El-Gohary (2013) proposed an IE method to extract design requirements from building codes to support automated building code checking. They used a building ontology to capture the semantic features in the building-code text, and developed a set of pattern-matching-based IE rules for the extraction. Second, Zhou and El-Gohary (2017) proposed an IE method to extract energy requirements from energy codes to support automated energy code checking. Compared with the first effort, they used a more detailed ontology to improve the extraction performance. They also proposed a combination of methods to deal with long provisions, hierarchical sentences, and exceptions.

## Knowledge Gaps

Despite the contributions of the aforementioned efforts, there is a lack of methods that can deal with the following three text complexities that characterize contract specifications: hierarchically complex text structures, incomplete sentence structures, and the variety of LODs. First, the text hierarchical structure in specifications is much deeper, wider, and more dynamic than the structure in energy codes in previous efforts (Zhou and El-Gohary 2017). The hierarchical complexity of the energy code is on the sentence level. However, specifications exhibit paragraph-level complexity, which is more challenging: the text is hierarchically deep and wide, as well as dynamic. An article in the specifications may consist of dozens of paragraphs, each paragraph may consist of up to 10 levels of subparagraphs, and each level may consist of dozens of subparagraphs. In contrast, a provision in the International Energy Conservation Code (IECC) usually has few levels of subprovisions, and each level has only several sentences. In addition, the paragraph-level complexities of specifications usually increase with project size. In contrast, different versions of the IECC tend to have similar complexity levels. Thus, automatically simplifying the text hierarchical structure in specifications for better identification of target requirements is more challenging.

Second, previous efforts (Zhang and El-Gohary 2013; Zhou and El-Gohary 2017) did not address text with incomplete sentence structures, which, unlike in codes, is common in specifications. IE from text with complete sentence structures is relatively easier because complete sentence structures have regular grammatical patterns. The experimental results of Zhou and El-Gohary (2017) showed that regular grammatical patterns usually implied strong dependency relationships among the target information, which was sufficient to reduce most of the text ambiguities. Text with incomplete sentence structures, lacking such regular patterns, thus is likely to suffer from weak dependency relationships that are insufficient to reduce ambiguities. Thus, there is a need for methods to deal with incomplete sentence structures to reduce ambiguities.

Third, existing methods are not able to recognize and differentiate the LODs of the information in the specifications. Unlike codes in previous efforts (Zhang and El-Gohary 2013; Zhou and El-Gohary 2017), specifications contain a wider range of requirements related to different LODs, in which requirements in irrelevant LODs are not related to checking of building energy design. For example, requirements in Part 2 of MasterFormat specifications not only relate to design of building elements, but also include fabrications, mixes, and factory finishing for those building elements (CSI and CSC 2009, 2017). Extraction of requirements in irrelevant LODs (i.e., information beyond the current or needed LOD) may result in potential compliance checking errors. Thus, there is a need for methods to correctly differentiate and identify requirements in the target LODs.

## Proposed Semantic Information Extraction Method for Automated Extraction of Energy Requirements from Contract Specifications

To address the aforementioned knowledge gaps, a semantic IE method for automated extraction of building energy requirements from contract specifications is proposed. Three text analytic methods are proposed and used to deal with the aforementioned text complexities.

### *Method 1: Proposed Domain-Specific Text Splitting and Stitching Method*

A domain-specific text splitting and stitching method is proposed to simplify the hierarchically complex text structures in specifications prior to IE. The proposed method uses a regular expressions-based pattern-matching technique to recognize automatically the text features that signal splitting, and to split the text to the most granular level. The proposed method includes two steps: splitting and stitching. First, each article is split to the lowest subparagraph level, with each resulting split text containing the text from a lowest-level subparagraph, each superlevel subparagraph, and the highest-level paragraph. Each resulting split text then is stitched with the numbers and titles of the corresponding section and article.

During splitting, the articles, paragraphs, and subparagraphs are recognized automatically based on the text feature patterns of their numbers and titles defined in the PageFormat, in which such numbers and titles signal the splitting. Regular expressions are used to define the patterns. During stitching, the numbers and titles of sections and articles are stitched for two reasons. First, the titles may

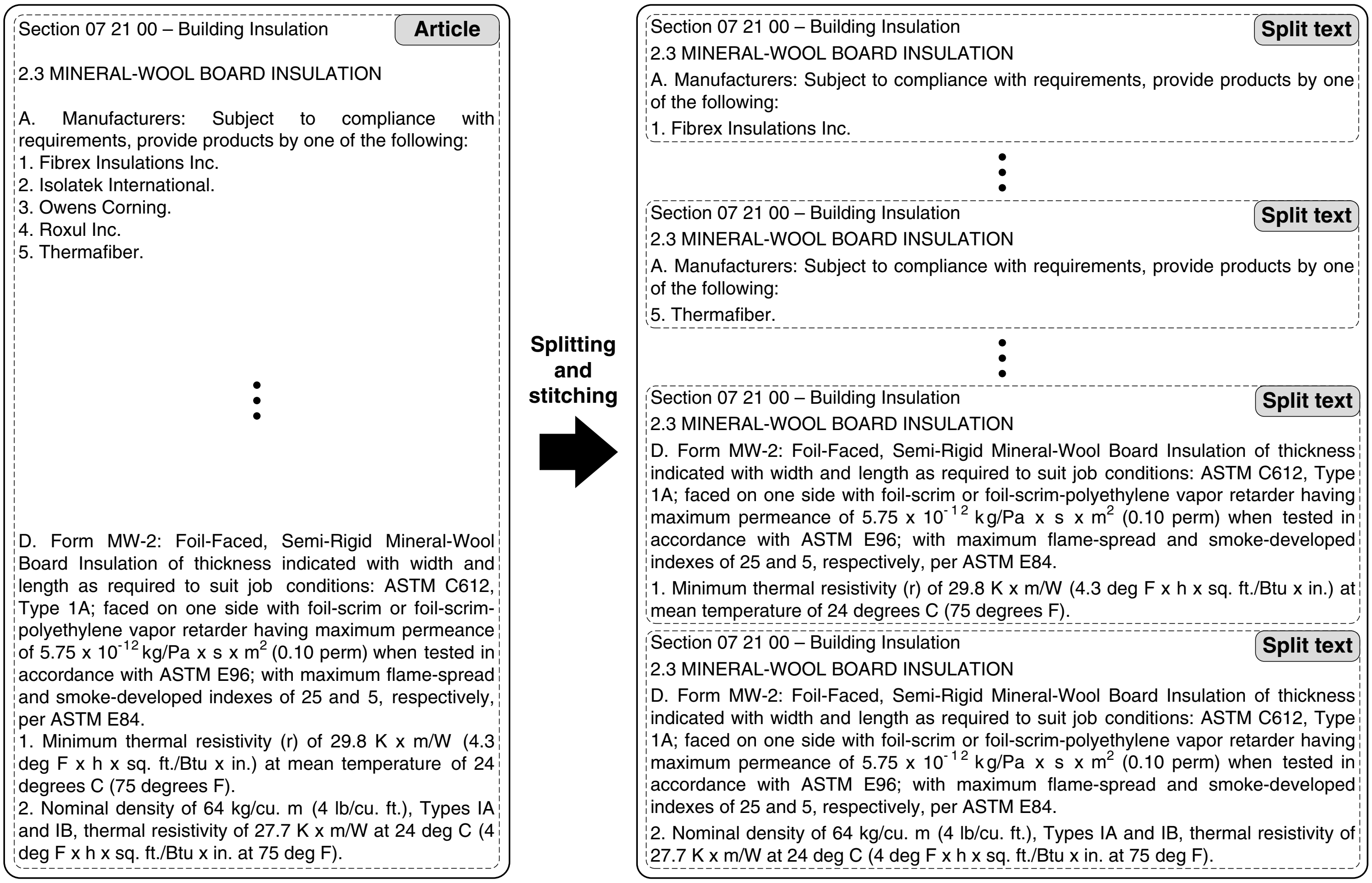


**Fig. 2.** Example illustrating the proposed domain-specific text splitting and stitching method.

contain target information that needs to be extracted. For example, the article title "MINERAL WOOL BOARD INSULATION" contains the target information "mineral wool board," which is an instance of the Subject of a requirement. Second, the numbers and titles are used as a reference for the requirement during the reporting of the compliance checking results. As an example, Fig. 2 shows the splitting and stitching for the article in Fig. 1.

### *Method 2: Proposed Incompleteness-Aware Sequential Dependency Extraction Method*

An incompleteness-aware sequential dependency extraction method is proposed to extract target information from text with incomplete sentence structures. The proposed method adapts the sequential dependency extraction method of Zhou and El-Gohary (2017) to text with incomplete sentence structures. The key of sequential dependency extraction is to use the dependency information among target information to assist in defining text feature patterns and developing extraction rules to reduce ambiguities and enhance extraction performance. The target information, in the context of ACC, is the nine semantic information elements (SIEs) of ACC (Zhang and El-Gohary 2013; Zhou and El-Gohary 2017): Subject, Subject Restriction, Compliance Checking Attribute, Deontic Operator Indicator, Quantitative Relation, Comparative Relation, Quantity Value, Quantity Unit/Reference, and Quantity Restriction. Detailed definitions of these SIEs were given by Zhang and El-Gohary (2013) and Zhou and El-Gohary (2017). Table 1 provides examples of SIE-represented energy requirements in a project's specifications. The dependency information is the conceptual dependency information among the SIEs that helps identify the extraction sequence of the SIEs, in which the use of dependee SIEs to extract a depender SIE reduces ambiguities.

The proposed method is novel because it further uses incompleteness features (i.e., features of incomplete sentence structures), along with dependency information, to help define the feature patterns and develop extraction rules that reduce ambiguities in text with incomplete sentence structures. Incompleteness features tend to cooccur with target information. The incompleteness features thus can reduce ambiguities that cannot be addressed solely by dependency information. For example, both "temperature" and "thermal resistivity" are potential candidates of the SIE Compliance Checking Attribute, but in the context of P1, only "thermal resistivity" (in bold) should be extracted as a Compliance Checking Attribute of a requirement; the "temperature" (in bold) refers to a testing condition. In Rule R1, the comma, which is an incompleteness feature that signals the co-occurrence of target information, reduces this ambiguity. It imposes an additional matching condition that a candidate Compliance Checking Attribute should be preceded by an incompleteness feature (in addition to being succeeded by Quantity Value and a Quantity Unit/Reference, the dependency information)

- P1: "C. Form MW-1: Un-faced, Semi-Rigid Mineral-Wool Board Insulation of thickness indicated with width and length as required to suit job conditions: ASTM C612, Type 1A; with maximum flame-spread and smoke-developed indexes of 15 and zero, respectively, per ASTM E84; passing ASTM E136 for combustion characteristics:
    1. Minimum thermal resistivity (r) of $29.8\ \mathrm{K} \times \mathrm{m/W}$ ($4.3\ \mathrm{deg\ F} \times \mathrm{h} \times \mathrm{sq.\,ft./Btu} \times \mathrm{in.}$) at mean **temperature** of 24 deg C (75 deg F).

**Table 1.** Examples of SIE-represented requirements in gold standard

| Split text | Semantic information element | Requirement 1 | Requirement 2 | Requirement 3 |
|---|---|---|---|---|
| Section 07 21 00—Building Insulation 2.3 MINERAL WOOL BOARD INSULATION | Subject | Foil-scrim or foil-scrim-polyethylene vapor retarder | Foil-faced semi-rigid mineral-wool board | Foil-faced semi-rigid mineral-wool board |
| D. Form MW-2: Foil-Faced, Semi-Rigid | Subject restriction | N/A | N/A | N/A |
| Mineral-Wool Board Insulation of thickness | Compliance checking attribute | Permeance | Nominal density | Thermal resistivity |
| indicated with width and length as required to | Deontic operator indicator | N/A | N/A | N/A |
| suit job conditions: ASTM C612, Type 1A; | Quantitative relation | N/A | N/A | N/A |
| faced on one side with foil-scrim or foil-scrim- | Comparative relation | Maximum | Greater than or equal[a] | Greater than or equal[a] |
| polyethylene vapor retarder having maximum permeance of $5.75 \times 10^{-12}$ kg/Pa × s × m$^2$ (0.10 perm) when tested in accordance with ASTM E96; with maximum flame-spread and smoke-developed indexes of 25 and 5, respectively, per ASTM E84. | Quantity value | $5.75 \times 10^{-12}$ (0.10) | 64 (4) | 27.7 (4) |
| 2. Nominal density of 64 kg/cu. m (4 lb/cu. ft.), thermal resistivity of 27.7 K × | Quantity unit/reference | kg/Pa × s × m$^2$ (perm) | kg/cu. m (lb/cu. ft.) | K × m/W (deg F × h × sq. ft./Btu × in.) |
| m/W at 24 deg C (4 deg F × h × sq. ft./Btu × in. at 75 deg F). | Quantity restriction | N/A | N/A | N/A |

[a]Greater than or equal was used as a default comparative relation if it is implicit in text.

2. Nominal density of 64 kg/cu. m (4 lb/cu. ft.), Types IA and IB, **thermal resistivity** of 27.7 K × m/W at 24 deg C (4 × deg F × h × sq. ft./Btu × in. at 75 deg F).

- R1: (Token.string == ",") + (potentialCommercialBuildingEnergyEfficiencyAttribute):cr + IN + QuantityValue + QuantityUnit/Reference –> cr.ComplianceCheckingAttribute.,

where the pattern (Token.string == ",") matches a comma, which is an incompleteness feature; potentialCommercialBuildingEnergyEfficiencyAttribute is a semantic feature that corresponds to a commercial building energy efficiency property (e.g., thermal resistivity), a concept in the ontology; IN is the POS tag for prepositions (e.g., of); and QuantityValue and QuantityUnit/Reference, the dependency information, match 27.7 (4) and K × m/W (deg F × h × sq. ft./Btu × in.), respectively.

### *Method 3: Proposed Detail-Aware Level of Development Extraction Method*

A detail-aware LOD extraction method is proposed to extract requirements that are in the target BIM LOD for compliance checking of BIMs. In this research, a LOD 350, as defined in the 2017 LOD specification (BIMForum 2017), is the target for both the requirements and the BIMs. The proposed method is novel because the target LOD is differentiated automatically based on analyzing the sentence grammatical moods in terms of syntactic text features. The proposed method uses an indirect way to extract information in LOD 350 from the text: (1) the extraction rules (Section "Extraction Rule Development Using Methods 2 and 3") are used to extract all information regardless of LOD; and (2) the information in LOD 400 or above then is recognized and removed from all the extracted information, leaving only information in LOD 350.

A rule-based pattern-matching approach is used to recognize the information in LOD 400 or above. As discussed in the section "Background," imperative sentences are used to convey information in LOD 400 or above, and the key feature of imperative sentences is an action verb. Therefore, a set of tagging rules is used to recognize and tag action verbs. Subsequently, sentences that begin with action verbs, except those in an exclusion list, are recognized as containing information in LOD 400 or above. Action verbs that are poor indicators of information in LOD 400 or above (such as "provide") are included in a list of exclusion verbs. POS tags are used to define the feature patterns in the tagging rules. Fig. 3 shows an example of a tagging rule, in which (1) the feature pattern is given in Lines 1–10, (2) the recognized action verbs are tagged with actionVerb in Line 12, and (3) the exclusion verbs are listed in Line 7. For example, this tagging rule was able to recognize that "hem" in S1 is an action verb, and accordingly S1 was removed

- S1: "Hem exposed edges of flashing on underside 0.013 m (1/2 in.)."

A new domain-specific POS tagger is proposed to support the recognition of action verbs, because a general POS tagger is limited in tagging domain-specific documents. In general, a POS tagger relies on two components to tag a word—a lexicon, and a ruleset. The lexicon is a dictionary of words in which each word has a list of related POS tags (i.e., a word may have multiple word classes). The first tag in the list is the most likely word class for that word in one type of document, and is the default tag assigned to a word. For example, "check NN VB VBP" is an item in the lexicon of the General Architecture for Text Engineering (GATE) Hepple POS Tagger (Cunningham et al. 2011), in which the related POS tags "NN VB VBP" indicate that the word "check" has three-word classes: it can be a singular or mass noun (NN, which is the default tag), a base form verb (VB), or a non-third-person singular present verb (VBP). The ruleset contains a set of conversion rules to convert the default tag to another tag if its adjacent words in the text match certain feature patterns. For example, Hepple's conversion rule "NN VB PREVTAG TO" (Cunningham et al. 2011) would convert the tag of "check" from its default NN to VB, if the previous word "check" in the text has the POS tag TO (i.e., if "check" is preceded by "to"). This rule still would fail to tag "check" correctly as a verb in cases in which it is not preceded by "to", such as in "Check the movement of the doors at both limits of travel." As this example shows, a general POS tagger such as Hepple's does not perform well in tagging action verbs because of missing domain-specific information in its lexicon and ruleset. For example, Hepple's lexicon is based on the text in the *Wall Street Journal* (Cunningham et al. 2011), in which "check" is more likely to be a noun referring to a written order directing a bank to pay money, and hence NN is its default tag. In contrast, in the context of contract specifications, "check" is more likely to be a verb meaning "verify." Therefore, the default for "check" in the new tagger is VB. The new domain-specific tagger was developed by building a domain-specific lexicon and ruleset based on Hepple's.



J. Comput. Civ. Eng., 2022, 36(5): 04022025

<table>
<tr><th>Line number</th><th>LOD tagging rule No. 1</th><th>Explanation</th></tr>
<tr><td>1:</td><td>(</td><td>Beginning of feature pattern;</td></tr>
<tr><td>2:</td><td>({Token.string == "."}|{Token.string == ":"})</td><td>The first token must be a period or a colon, which is an incompleteness feature;</td></tr>
<tr><td>3:</td><td>({CC})?</td><td>The next token may be an optional coordinating conjunction;</td></tr>
<tr><td>4:</td><td>({Token.string == "do"}{Token.string == "not"})?</td><td>The next two tokens may be an optional phrase "do not";</td></tr>
<tr><td>5:</td><td>({RB})?</td><td>The next token may be an optional adverb;</td></tr>
<tr><td>6:</td><td>({NN})?</td><td>The next token may be an optional singular or mass noun;</td></tr>
<tr><td>7:</td><td>(({VB, Token.string != "provide", Token.string != "include", Token.string != "refer", Token.string != "comply", Token.string != "conform", Token.string != "furnish", Token.string != "see", Token.string != "use", Token.string != "do", Token.string != "be", Token.string != "pass", !VB within paragraphTitle} |<br>{VBP, Token.string != "provide", Token.string != "include", Token.string != "refer", Token.string != "comply", Token.string != "conform", Token.string != "furnish", Token.string != "see", Token.string != "use", Token.string != "do", Token.string != "be", Token.string != "pass", !VBP within paragraphTitle})):tempTag</td><td>The candidate action verb must<br>(1) be a base form verb or a non-3rd person singular present verb; and<br>(2) not be in the list of exclusion verbs: "provide, include, refer, comply, conform, furnish, see, use, do, be, pass"; and<br>(3) not be part of a paragraph title.<br>A tag "tempTag" is assigned to each candidate;</td></tr>
<tr><td>8:</td><td>{Token.kind != punctuation, Token.category != MD, !paragraphNumber}</td><td>The token immediately after a candidate must not be a punctuation, modal verb or paragraph number;</td></tr>
<tr><td>9:</td><td>({Token, Token.string != ":", Token.category != MD, Token.string != "are", Token.string != "is", !paragraphNumber})[4]</td><td>Any of the next four consecutive tokens must not be a colon, modal verb, "are", "is", or paragraph number;</td></tr>
<tr><td>10:</td><td>)</td><td>End of feature pattern;</td></tr>
<tr><td>11:</td><td>--></td><td>The boundary of feature pattern;</td></tr>
<tr><td>12:</td><td>:tempTag.actionVerb = {rule = "LOD tagging rule No. 1"}</td><td>The candidates that match the feature pattern are action verbs, and are tagged as "actionVerb";</td></tr>
</table>

**Fig. 3.** Example of a tagging rule for recognizing action verbs.

## Implementation of Proposed Semantic Information Extraction Method

The proposed semantic IE method was implemented and tested in extracting building energy requirements from the contract specifications of an educational building project. The scope of testing was limited to two subtopics of commercial building energy efficiency: thermal insulation, and lighting power. Only text from "Part 2 Products" of the specifications was used in testing because that part contains all related design requirements for compliance checking. The implementation and testing included seven primary steps (Fig. 4).

### *Domain-Specific Text Classification*

Domain-specific text classification was conducted to filter out articles that were not related to building energy efficiency. Fig. 5 shows an example of an irrelevant article that was filtered out, because it prescribes fireproofing requirements. The TC method proposed by Zhou and El-Gohary (2016) was used in this research to classify the articles. Details of the TC method were given by Zhou and El-Gohary (2016).

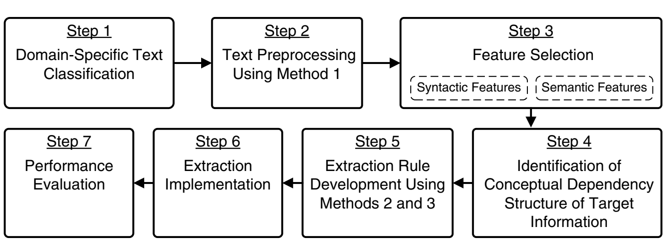


**Fig. 4.** Implementation of proposed semantic information extraction method.

### *Text Preprocessing Using Method 1*

The proposed domain-specific text splitting and stitching method was implemented first to deal with the hierarchically complex text structures. The proposed method was used to split the classified articles. For example, an article with number and title "2.3 MINERAL WOOL BOARD INSULATION" was classified as thermal insulation topic (Fig. 2). The article includes four paragraphs, of which Paragraphs A and D include five and two subparagraphs, respectively. The article was split into the subparagraph level, in which each split text included the text from the subparagraph and paragraph, and was stitched with the numbers and titles of the corresponding section (Section 07 21 00—Building Insulation) and article (2.3 MINERAL WOOL BOARD INSULATION). The resulting split text then was preprocessed using three common text processing techniques: tokenization, sentence splitting, and morphological analysis.

Tokenization aims to split the raw text into tokens (e.g., words, numbers, punctuations, symbols, and whitespace) to identify

2.2 SPRAYED FIRE-RESISTIVE MATERIALS
A. SFRM: Manufacturer's standard, factory-mixed, lightweight, dry formulation, complying with indicated fire resistance design, and mixed with water at Project site to form a slurry or mortar before conveyance and application.
1. Products: Subject to compliance with requirements, provide one of the following:
a. Grace, W. R. & Co. - Conn.; Grace Construction Products; Monokote MK-6 Series.
b. Isolatek International; Cafco 300.
c. Southwest Fireproofing Products Co.; Type 5GP.

•
•
•

6. Combustion Characteristics: ASTM E136.
7. Surface-Burning Characteristics: Comply with ASTM E84; testing by a qualified testing agency. Identify products with appropriate markings of applicable testing agency.
a. Flame-Spread Index: 10 or less.
b. Smoke-Developed Index: 10 or less.

**Fig. 5.** Example of an irrelevant article.

sentence boundaries and to prepare for POS tagging (Manning and Schutze 1999; Moens 2006). For example, the text "2. Nominal density of 64 kg/cu. m." was tokenized into "'2' '.' 'Nominal' 'density' 'of' '64' 'kg' '/' 'cu' '.' 'm' '.'" (whitespace tokens are not shown).

Sentence splitting aims to split the text into individual sentences based on the sentence boundary indicators (e.g., periods, exclamation marks, and question marks) (Jurafsky and Martin 2020). Sentence boundary indicators in contract specifications have ambiguities similar to those in energy codes. For example, the periods in the subparagraph number "2." and in the measurement unit "lbs. per cubic foot" are not sentence boundary indicators. Therefore, a set of domain-specific sentence splitting rules was developed manually to correctly recognize and split sentences in specifications.

Morphological analysis aims to convert the words in derivational (e.g., affixes such as -ly and -ion) or inflectional forms (e.g., plural or progressive) to their base form (Manning and Schutze 1999). For example, both "limiting" and "limiter" were converted to the base form "limit." Morphological analysis helped in recognizing and selecting the semantic features (section "Feature Selection") by mapping the morphologically analyzed text to the ontology concepts. For example, through morphological analysis, "current limiting circuit breakers" in the natural text was mapped to the concept "current limit circuit breaker" in the ontology. The concept then was used as a semantic feature.

### *Feature Selection*

The target information is recognized based on the features of the text (section "Extraction Rule Development Using Methods 2 and 3"). Two types of features were used: syntactic and semantic features.

#### Syntactic Features

Three types of syntactic features were used: POS tags, gazetteers, and domain-specific tags. POS tagging aims to assign a POS tag to each word based on its word class (e.g., adjective, noun, preposition) (Moens 2006). For example, the tag JJ was assigned to adjectives (e.g., "thermal"). In this research, a domain-specific POS tagger was developed and used (section "Method 3: Proposed Detail-Aware Level of Development Extraction Method"). Each POS tag was used as a syntactic feature.

A gazetteer refers to a list of words belonging to the same category (e.g., country names) (Wimalasuriya and Dou 2010). In this research, two gazetteers were developed manually: a comparison gazetteer, and a measurement unit gazetteer. The first included words and phrases that indicate comparison relationships (e.g., "exceed" and "less than"). The second included words and symbols that represent quantity units (e.g., "watt" and "K × m/W"). Each gazetteer was assigned a tag, and each tag was used as a syntactic feature.

Fourteen domain-specific tags were defined in this research, and a set of tagging rules was developed to tag the text with these tags. An example of such tags is "domainSpecificCD," which refers to domain-specific cardinal numbers (e.g., "1-1/8"). Each tag was used as a syntactic feature.

#### Semantic Features

The semantic features of the text were captured using the commercial building energy ontology (Zhou and El-Gohary 2017), which covers concepts that are related to commercial building energy efficiency. A partial view of the energy ontology is shown in Fig. 6. The web ontology language in-memory (OWLIM) Ontology Editor of GATE (Cunningham et al. 2011), a text processing platform, was used to build the ontology. The ontology was processed by the GATE OntoRoot Gazetteer module to create a concept gazetteer and to parse the hierarchical is-a relationships among the concepts. Each concept in the gazetteer was used as a semantic feature. The is-a relationships helped recognize subconcept relationships for the extraction (section "Extraction Rule Development Using Methods 2 and 3").

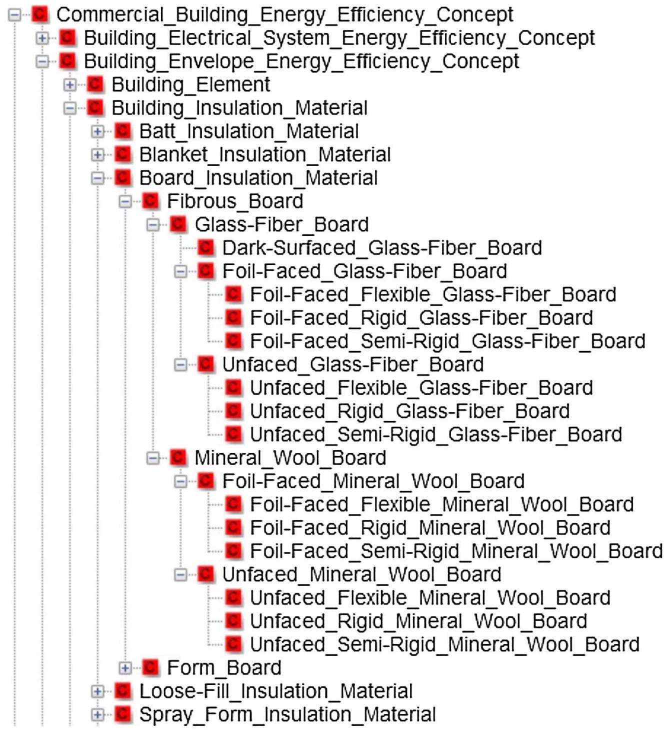


**Fig. 6.** Partial view of the commercial building energy ontology.

### *Identification of Conceptual Dependency Structure of Target Information*

A conceptual dependency structure (CDS) was developed to represent the dependency information among the target information—the nine SIEs (section "Method 2: Proposed Incompleteness-Aware Sequential Dependency Extraction Method"). Such dependency information helps to identify the extraction sequences of the SIEs. The CDS of Zhou and El-Gohary (2017) was adapted; the text in contract specifications was analyzed to identify the dependency relationships among the SIEs, and the CDS of Zhou and El-Gohary (2017) was modified accordingly. Fig. 7 shows the developed CDS for specifications, in which arrows represent the dependency relationships and numbers indicate the extraction sequence. The CDS for specifications differs from the CDS of Zhou and El-Gohary (2017) in two main ways. First, it contains 50% more dependency relationships among all SIEs, in which most of these extra relationships involve two SIEs—Deontic Operator Indicator, and Quantitative Relation. For example, Deontic Operator Indicator and Quantitative Relation depend on two new SIEs—Quantity Value, and Compliance Checking Attribute. Second, Subject is moved up in the CDS, before Deontic Operator Indicator, indicating that Deontic Operator Indicator is no longer used as a dependency information in the extraction of Subject. These two dependency differences are due to the differences in sentence structures and writing style across codes and specifications (as discussed in the section "Background"). For example, modal verbs (which are instances of Deontic Operator Indicator) are less likely to be used in specification sentences

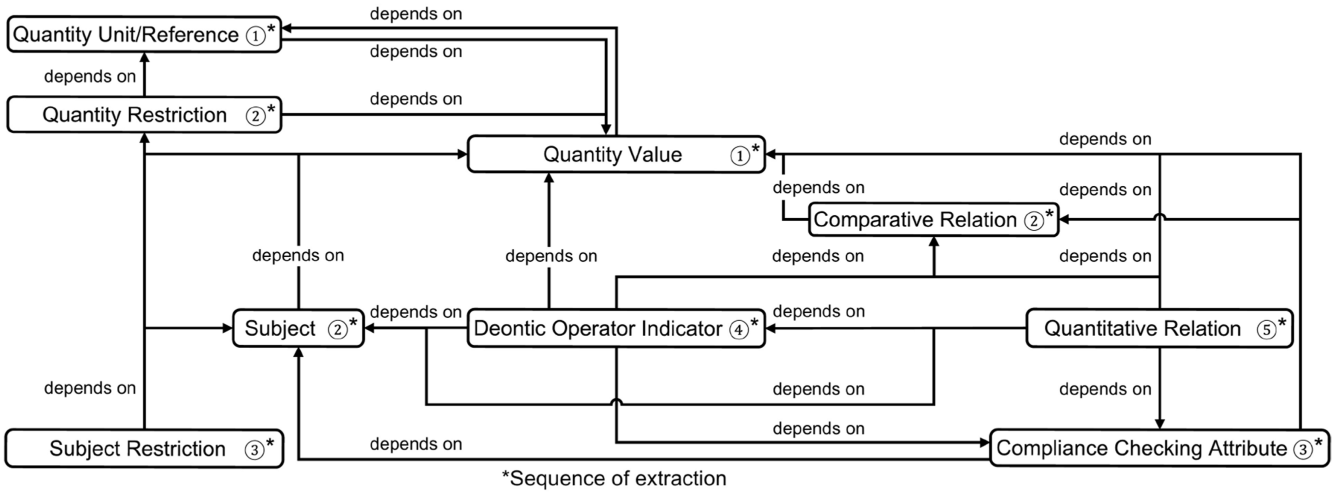


**Fig. 7.** Conceptual dependency structure.

(e.g., the modal verb "shall" is replaced by a colon in streamlined writing), which results in the need to use other information (e.g., incompleteness features) to extract Subjects.

### *Extraction Rule Development Using Methods 2 and 3*

The extraction rules were developed manually after analyzing the text feature patterns in a sample of text (called development data). The development data included 400 sample sentences from five contract specifications which cover five energy topics (air leakage, thermal insulation, fenestration, lighting system control, and lighting power). Each extraction rule has two sides: the left side models the feature patterns using regular expressions, and the right side indicates the target information that needs to be extracted. For example, R2 extracted "permeance" from S2, as an instance of Compliance Checking Attribute. The rule states that if a commercial building energy efficiency attribute is preceded by a Comparative Relation and is followed by IN, Quantity Value, and Quantity Unit/Reference, the attribute should be extracted as an instance of Compliance Checking Attribute

- S2: "Faced on one side with foil-scrim or foil-scrim-polyethylene vapor retarder having maximum permeance of $5.75 \times 10^{-12}$ kg/Pa $\times$ s $\times$ m$^2$ (0.10 perm) when tested in accordance with ASTM E96."
- R2: ComparativeRelation + (potentialCommercialBuilding EnergyEfficiencyAttribute):cr + IN + QuantityValue + Quantity Unit/Reference –> cr.ComplianceCheckingAttribute., where potentialCommercialBuildingEnergyEfficiencyAttribute is a semantic feature that matches a concept in the ontology; IN is a POS tag that matches prepositions such as of; Comparative Relation, QuantityValue, and QuantityUnit/Reference are three SIEs that are used as dependency information (matching "maximum", "0.10", and "perm", respectively, in S2); and cr is the pointer set to the semantic feature potentialCommercial BuildingEnergyEfficiencyAttribute.

During the development of extraction rules, the proposed incompleteness-aware sequential dependency extraction method (Method 2) and detail-aware LOD extraction method (Method 3) were considered and implemented to deal with incomplete sentence structures and the variety of LODs. For Method 2, the incompleteness features were used as text features, along with the dependency information among target SIEs, to define the text feature patterns and develop the extraction rules (Rule R1). For Method 3, the tagging rules (section "Method 3: Proposed Detail-Aware Level of Development Extraction Method") were developed and implemented along with the extraction rules to recognize the action verbs. The split sentences (section "Text Preprocessing Using Method 1") that included the action verbs were recognized as imperative sentences. The imperative sentences, from which all information was extracted by the extraction rules, then were removed. In the LOD extraction, (1) 71 domain-specific words, with their related POS tags, and 6 conversion rules were added to the lexicon and ruleset of the GATE Hepple POS Tagger; and (2) 5 tagging rules were developed to recognize the action verbs of imperative sentences.

To address conflicts that may arise in the extraction, 12 domain-specific conflict resolution (CR) rules were developed. These rules can be categorized into two types—global and local CR rules. Global rules are applicable to all SIEs, whereas local rules are applicable only to specific SIEs. For example, CR1 is a global rule, which applies to all SIEs. It extracted "2.0" (rather than "0") from S3, as an instance of Quantity Value, because "2.0" is the longest matching instance. In contrast, CR7 is a local rule, which applies to Subjects only. It extracted the more-specific "fluorescent electronic ballast" (rather than "ballast") from S4, as an instance of Subject

- S3: "Sprayed Polyurethane Foam Sealant: 1- or 2-component, foamed-in-place, polyurethane foam sealant, 24 to 32 kg/cu. m (1.5 to 2.0 lb/cu. ft.) density."
- CR1: "If there are multiple instances that match the text feature patterns, only the longest matching instance should be extracted."
- S4: "2.9 FLUORESCENT ELECTRONIC BALLAST F. Ballast Requirements: 5. Ballast power factor shall be greater than 95%."
- CR7: "If there are multiple matching instances for Subject, the instance that corresponds to the lower-level concept (i.e., the more-specific concept in the ontology) should be extracted."

### *Extraction Implementation*

The proposed semantic IE method was implemented in a Java-based platform. The application programming interfaces (APIs) of the following modules in the a nearly-new IE (ANNIE) system of GATE (Cunningham et al. 2011) were used to implement the methods discussed in the aforementioned sections, including the ANNIE English Tokeniser, ANNIE Sentence Splitter, GATE Morphological Analyser, ANNIE POS Tagger, ANNIE Gazetteer, OntoRoot Gazetteer, and JAPE Transducer. Each of these modules may have a set of initialization parameters, and each parameter was set experimentally in this research. For example, the values of the parameters lexiconURL and rulesURL in the ANNIE POS Tagger were set to the developed domain-specific lexicon and ruleset. The two gazetteers (section "Feature Selection") were added to the ANNIE Gazetteer. The domain-specific sentence splitting rules (section "Text Preprocessing Using Method 1"), the tagging rules for the 14 domain-specific tags (section "Feature Selection"), the tagging rules for recognizing the action verbs in the detail-aware LOD extraction method (section "Extraction Rule Development Using Methods 2 and 3"), and the extraction rules (section "Extraction Rule Development Using Methods 2 and 3") were developed in the grammar of Java Annotation Patterns Engine (JAPE) (Cunningham et al. 2011) using the JAPE editor Vi improved (Vim) (Robbins et al. 2008), and were added to the JAPE Transducer. The domain-specific text classification (section "Domain-Specific Text Classification"), the domain-specific text splitting and stitching (section "Text Preprocessing Using Method 1"), and the CR rules (section "Extraction Rule Development Using Methods 2 and 3") were implemented in separate Java programs.

### Performance Metrics

Recall and precision were used to measure the performance (Maynard et al. 2006; Moens 2006). Recall refers to the percentage of the total number of correctly extracted instances out of the total number of instances in the gold standard. Precision refers to the percentage of the total number of correctly extracted instances out of the total number of extracted instances. The confidence interval ($p$) was calculated for recall and precision to test the statistical significance of the results (Goutte and Gaussier 2005). The Wilson score without continuity correction (Wilson 1927) was used to measure the confidence intervals because it is both computationally simple and satisfactory (Newcombe 1998). It was calculated using Eq. (1), where $p_0$ refers to the values of precision or recall, $q_0 = 1 - p_0$, $\lambda$ is the critical value for the confidence interval, $n$ refers to either the total number of extracted instances (for calculating $p$ for precision) or the total number of instances in the gold standard (for calculating $p$ for recall), and $t = \lambda^2/n$

$$p = \frac{p_0 + t/2}{1+t} \pm \frac{\sqrt{p_0 q_0 t + t^2/4}}{1+t} \qquad (1)$$

## Experimental Results and Analysis

### Experimental Setup

A testing data set was used to evaluate the performance. The data set was prepared based on the contract specifications of an educational building project in Illinois, which were different from the five specifications that were used to prepare the development data (section "Extraction Rule Development Using Methods 2 and 3"). The sections from Divisions 07 and 26 of the specifications were selected because they contain thermal-insulation and lighting-power requirements. A total of 148 requirements were selected randomly from the 393 requirements in these sections, which formed the testing data set. A gold standard was developed for performance evaluation. A gold standard includes the human-annotated testing data, annotated with the target information. The extraction results then are compared to the gold standard to evaluate precision and recall. To develop the gold standard, the requirements were annotated by three annotators—the first author, and two other researchers. The $F$-measure was used to measure the reliability of the annotated gold standard (the equation of the $F$-measure is given by Pestian et al. 2012). An initial interannotator agreement of 88% for the $F$-measure was achieved, which is considered to be a sufficient score. "An $F$-measure of 0.8 or above is generally considered sufficient inter-annotator agreement" (Pestian et al. 2012). The disagreements among annotators were resolved to achieve final full annotator agreement using the common majority voting technique (e.g., Vieira and Poesio 2000; Klebanov and Beigman 2009): (1) if the majority (i.e., at least two) of the annotators achieved agreement, then the agreed-on annotation was used, and (2) if the majority (i.e., at least two) of the annotators did not achieve agreement, then a discussion was conducted until they achieved agreement and the agreed annotation was used. Because of the unambiguous nature of quantitative requirements within the scope of the authors' research, along with the well-defined SIEs used in the proposed method, there was a majority agreement in formulating the gold standard. As an illustration, Table 1 lists three SIE-represented requirements from the gold standard, which were from one article after domain-specific text splitting and stitching in Fig. 2.

### Performance Results

The experimental results are summarized in Table 2. A total of 99 extraction rules were developed. The gold standard contained a total of 787 instances for all 9 SIEs. The calculation of precision and recall for those 9 SIEs followed the most conservative and stringent rule—only an exact match was considered to be correct. An example of calculating the precision and recall by comparing the extraction results with the gold standard is provided in Table 3. An overall performance of 96.8% recall [with a confidence interval of (95.4%, 97.8%) at the 95% confidence level] and 97.6% precision [with a confidence interval of (96.2%, 98.4%) at the 95% confidence level] was achieved. This promising performance indicates that the proposed semantic IE method is successful in extracting building energy requirements from contract specifications.

**Table 2.** Experimental results of extracting building energy requirements from contract specifications

| Parameter/measure | Subject | Subject restriction | Compliance checking attribute | Deontic operator indicator | Quantitative relation | Comparative relation | Quantity value | Quantity unit/reference | Quantity restriction | Total | Average |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Information extraction rules | 20 | 4 | 11 | 4 | 9 | 6 | 20 | 20 | 5 | 99 | — |
| Instances in gold standard | 148 | 35 | 108 | 15 | 29 | 148 | 148 | 148 | 8 | 787 | — |
| Instances extracted | 137 | 33 | 107 | 15 | 29 | 147 | 156 | 149 | 8 | 781 | — |
| Instances correctly extracted | 132 | 33 | 104 | 15 | 29 | 147 | 147 | 147 | 8 | 762 | — |
| Precision (%) | 96.4 | 100.0 | 97.2 | 100.0 | 100.0 | 100.0 | 94.2 | 98.7 | 100.0 | — | 97.6 |
| Recall (%) | 89.2 | 94.3 | 96.3 | 100.0 | 100.0 | 99.3 | 99.3 | 99.3 | 100.0 | — | 96.8 |

**Table 3.** Example comparing extraction results with gold standard for performance calculation

| Split text | Semantic information element | Gold standard | Extraction result | Comparison and performance result |
|---|---|---|---|---|
| 072100 Building Insulation 2–PRODUCTS 2.2 GLASS-FIBER BOARD INSULATION D. Form FG-7: Semi-Rigid, Glass-Fiber Board Insulation with white faced on one side All Service Jacket (ASJ), of thickness indicated with width and length as required to suit job conditions: For application on underside of exposed floor decks, ASTM C 612, Type 1A & 1B. | Subject | Semi-rigid glass-fiber board | Glass-fiber board | Incorrect (recall error + precision error) |
| | Subject restriction | — | — | — |
| | Compliance checking attribute | Thermal resistivity | Thermal resistivity | Correct |
| | Deontic operator indicator | — | — | — |
| | Quantitative relation | — | — | — |
| 3. Nominal density of 48 kg/cu. m (3 lb/cu. ft.), thermal resistivity of 29.8 K × m/W at 24 deg C (4.3 deg F × h × sq. ft./Btu × in. at 75 deg F). | Comparative relation | Greater than or equal[a] | Greater than or equal[a] | Correct |
| | Quantity value | 29.8 (4.3) | 29.8 (4.3) | Correct |
| | Quantity unit/reference | K × m/W (deg F × h × sq. ft./Btu × in.) | K × m/W (deg F × h × sq. ft./Btu × in.) | Correct |
| | Quantity restriction | — | — | — |

[a]Greater than or equal was used as a default comparative relation if it is implicit in text.

### *Effects of Incompleteness-Aware Sequential Dependency Extraction*

The experimental results show the effectiveness of the incompleteness features along with the dependency information in two ways. First, the use of incompleteness features improved the overall extraction performance, by 2.1% (from 95.5% to 97.6%) precision and 0.5% (from 96.3% to 96.8%) recall. Second, the incompleteness features were effective in reducing ambiguities. Dependency information alone may not be sufficient. For example, in Fig. 8, the letter A (in bold), which is a potential instance of Quantity Unit/Reference, may create ambiguity. It may be a paragraph number (e.g., "A. Item 1: Busway:"), a part of a designation number (e.g., "ASTM C612, Type 1A & 1B."), or the abbreviation of ampere (e.g., "C. Auxiliary contacts, rated 10 A, 250 VAC,"), but only in the last case should it be extracted as an instance of Quantity Unit/Reference. The proposed method was able to avoid such ambiguities using incompleteness features, in addition to the dependency information.

The analysis of errors that were reduced by using incompleteness features indicates two interesting findings. First, incompleteness features may have different contributions to the extraction of different SIEs. This is reflected by the fact that SIEs that are on top of the CDS tended to achieve the most performance improvement. For example, the SIE Quantity Value improved by 3.7% (from 90.5% to 94.2%) recall and 2.7% (from 96.6% to 99.3%) precision; the Quantity Unit/Reference improved by 3.0% (from 95.7% to 98.7%) precision; the Subject improved by 4.1% (from 92.3% to 96.4%) precision; and the Compliance Checking Attribute improved by 0.9% (from 96.3% to 97.2%) precision. Second, the incompleteness features may help recognize the boundary (beginning or ending) of target information within the text to reduce errors. For example, the incompleteness feature "," in S5 signals the beginning of target information. Thus, "10", rather than "4397, 10", correctly was extracted as a Quantity Value. Similarly, because of the lack of incompleteness features in S6 to signal the boundary, "2.5"

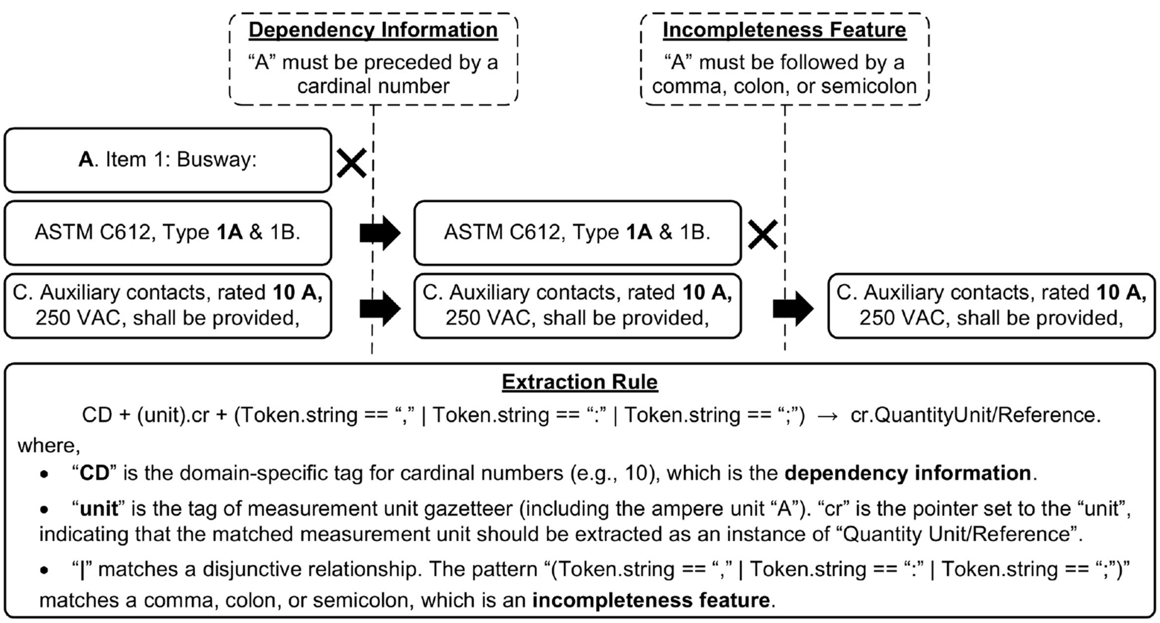


**Fig. 8.** Example of using incompleteness features along with dependency information in reducing ambiguities.

and “INSULATION”, although matching the feature patterns, correctly were not extracted as a Quantity Value and a Compliance Checking Attribute, respectively

- S5: “Polyethylene film: ASTM D 4397, $2.5 \times 10^{-4}$ m (10 mils) thick, . . . .”
- S6: “071326 2-PRODUCTS 2.5 INSULATION A. Board Insulation: Extruded-polystyrene insulation . . . .”

### Effects of Detail-Aware LOD Extraction

The experimental results show that no extraction errors were caused by the detail-aware LOD extraction, which indicates the success of the proposed method in extracting information in LOD 350. However, a few errors occurred in tagging the action verbs for two reasons. First, the incomplete sentence structures make it challenging to tag words that have word-class ambiguity. As discussed in the section “Method 3: Proposed Detail-Aware Level of Development Extraction Method,” (1) some words may have multiple word classes, which may result in word-class ambiguity, and (2) the POS tagger relies on the context of a word (i.e., its adjacent words in text) to determine its word class. However, text with incomplete sentence structures may lack sufficient contextual information to determine the correct word class, which may result in POS tagging errors. For example, the word “safeguard” has word-class ambiguity; it could be either a noun or a verb. It was tagged incorrectly as an action verb in T1 because it is the first word of the sentence and has only partial context information (e.g., “Gravel,” “Stop,” and “System”) to determine its word class. A semantic POS tagger could have helped address this word-class ambiguity; it could have used domain-specific knowledge (e.g., in the form of ontology or gazetteer) to disambiguate the word classes. For example, “Safeguard” is the trademark name for a gravel stop system, and therefore was more likely to be a noun:

- T1: “a. Safeguard Gravel Stop System by W.W. Hickman Co.”

Second, a few verbs that were included in the verb exclusion list occasionally were used in imperative sentences that prescribe requirements in LOD 400 or above. For example, the verb “provide” was added to the verb exclusion list, but it should have been tagged as an action verb in T2, because the imperative sentence contains a fabrication requirement for flat lock seams (i.e., a requirement in LOD 400 or above)

- T2: “C. Provide flat lock seams, except corners. Fabricate corners minimum $0.45\ \mathrm{m} \times 0.45\ \mathrm{m}$ (18 in. × 18 in.) mitered and sealed as one piece.”

### Error Analysis

An error analysis was conducted to identify the sources of extraction errors. Six sources of errors were identified: dependency information, uncommon patterns, conflict resolution errors, extraction tool errors, ambiguous concept representation, and coreference ambiguity. First, missing the extraction of the dependee may result in missing the extraction of the depender. For example, in T3, “maximum” was not extracted as a Comparative Relation, because its dependees, “4” (Quantity Value) and “′” (the unit symbol for feet, a Quantity Unit/Reference), were not extracted

- T3: “2. Board size: 4′-0″ × 4′-0″ maximum.”

Second, uncommon patterns may result in missing the extraction of target information. Compared with energy codes, uncommon patterns are more likely to occur in contract specifications due to their special text representations. For example, “4” and “′” were not extracted as instances of Quantity Value and Quantity Unit/Reference from T3 because of this rare pattern (“4′-0″ × 4′-0″”) for representing values and units.

Third, conflict resolution errors were due to missing a CR rule or due to an error caused by a CR rule. Missing a CR rule may result in extracting irrelevant information. For example, in T4, four instances of Quantity Value [i.e., “$5.75 \times 10^{-12}$ (0.10)”, “25”, “36 (2.25)”, and “29.8 (4.3)”] and three instances of Quantity Unit/Reference [i.e., “$\mathrm{kg/Pa \times s \times m^2}$ (perm)”, “kg/cu. m (lb/cu. ft.)”, and “K × m/W (deg F × h × sq. ft./Btu × in.)”] were extracted, in which “25” was incorrect, because it refers to the flame spread index, an attribute for an irrelevant fire safety requirement. The current CR rules were not able to deal with such errors. On the other hand, errors caused by the CR rules may result in missing the extraction of target information. For example, CR10 caused an extraction error for T5. “Impedance” incorrectly was deleted from T5 because it is contained in “impedance tolerance”. Both “impedance” and “impedance tolerance” should have been extracted as Compliance Checking Attribute instances, because each refers to a requirement

- T4: “C. Form FG-4: Foil-Faced, Glass-Fiber Board Insulation of thickness indicated with width and length as required to suit job conditions: ASTM C612, Type IA; faced on one side with foil-scrim-kraft or foil-scrim-polyethylene vapor retarder having maximum permeance of $5.75 \times 10^{-12}\ \mathrm{kg/Pa \times s \times m^2}$ (0.10 perm) when tested in accordance with ASTM E96, with maximum flame-spread and smoke-developed indexes of 25 and 50, respectively, per ASTM E84.

  1. Nominal density of 36 kg/cu. m (2.25 lb/cu. ft.), thermal resistivity of 29.8 K × m/W at 24 deg C (4.3 deg F × h × sq. ft./Btu × in. at 75 deg F).”
- T5: “H. Impedance: Minimum 5.75 percent. The impedance tolerance shall be plus or minus 7.5 percent.”
- CR10: “If there are two instances for the SIE Compliance Checking Attribute and one instance is contained in another, keep the longer instance and delete the shorter one.”

Fourth, few errors were caused by the extraction tool errors. For example, in T6, “time delay” was not extracted as a Compliance Checking Attribute because the GATE Morphological Analyzer failed to recognize the semantic feature “time delay”

- T6: “g. When required by National Electrical Code or indicated on Project Drawings, the control system shall include a ground fault monitoring relay. The relay shall be adjustable from 100 to 1,200 amps, and include adjustable time delay of 0–1.0 s.”

Fifth, a concept may have an ambiguous representation in the text, which may result in failure to recognize its semantic features and consequently failure of the extraction. For example, in T7, the concept “recessed mounted panelboard” was represented as “Panelboard, recessed mount,” which resulted in failure to recognize the semantic feature and extract “Panelboard, Recessed Mount” as a Subject instance. This indicates that further processing may be required, in addition to morphological analysis, to help recognize the semantic features of the text with ambiguous concept representation

- T7: “1. Panelboard, recessed mount, 208/120 volt, 3 phase, 4 wire, S/N, ground bus, copper bus, bolt-on breakers, NEMA 1 enclosure, see schedules for size and configuration, see Specifications for additional information.”

Sixth, coreference ambiguity may result in the extraction of inaccurate information. For example, in T6, “relay” incorrectly was extracted as an instance of Subject, which is inaccurate—because “relay” actually refers to “ground fault monitoring relay” in a previous sentence. Coreference ambiguity was out of the research scope.



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## Limitations and Future Work

Four limitations related to the scope of the work and the testing are acknowledged. First, the proposed IE method is limited to the extraction of requirements from MasterFormat specifications. However, the promising performance achieved is not specific or limited to the format of MasterFormat. Different types of documents have different text characteristics (e.g., terminologies, sentence structures, text features, and grammars). Each characteristic could pose a benefit for or challenge to natural language processing and document analytics. Leveraging the benefits and overcoming the challenges depends on the success of the processing and analytics methods used. For example, the incomplete sentence structures of the MasterFormat pose a challenge because they hinder the ability to benefit from full sentence analysis and dependency information, which are addressed successfully by the proposed method through capturing the regularity of incompleteness text feature patterns. In future research, the proposed method could be tested on other specification standards such as the UniFormat (CSI 2021). We expect that the proposed method will generalize well to other formats of specifications, with necessary adaptions, because different types of specifications share common text characteristics, with some expected differences. For example, the UniFormat shares some common characteristics with the MasterFormat (e.g., incomplete sentence structures and streamlined writing style), but with some differences in the level of detail of the text. New text complexities that could arise can be addressed through adaptations and/or extensions of the proposed methods.

Second, the proposed detail-aware LOD extraction method is limited to the extraction of information in LOD 350, because the research scope currently is limited to checking the compliance of BIMs in LOD 350. In future research, to extend the proposed IE method to multiple LODs, machine-learning techniques could be used to develop a text classifier to classify the sentences automatically according to the required LOD, in which the grammatical moods, types of action verbs, and paragraph headings could be selected as features for the machine learning.

Third, the proposed method was tested only in extracting two types of requirements—thermal insulation, and lighting power. In future work, the proposed method could be tested on more energy topics (e.g., fenestration), as well as additional types of topics (e.g., safety). The methods also potentially are applicable to other types of documents that may share similar characteristics (e.g., hierarchically complex text structures) with specifications [e.g., Occupational Safety and Health Administration (OSHA) standards] for use in other ACC areas. Minor extension effort would be needed in this case. For example, the commercial building energy ontology may need to be extended if the new energy subtopics are not already covered in the scope of the ontology. The lexicon and the ruleset of the domain-specific tagger may need to be extended to cover new domain-specific words and conversion rules. In addition, the extraction rules and tagging rules (section "Extraction Implementation") also may need some modification or extension. The rules could be either reused as is or adapted (modified or extended) based on additional development text. Compared with the initial efforts, future adaptation efforts should require significantly less work. A similar level of performance is expected after such necessary extension is conducted.

Fourth, due to the manual effort involved in its development, the gold standard included only 148 requirements from the contract specifications of one project. In future work, the authors plan to test the proposed method further using a larger set of requirements from the specifications of multiple projects. A comparable level of performance is expected.

In addition, in ongoing and future work, the authors will test and compare the use of rule-based and machine-learning approaches in the extraction of requirements from specifications. This could provide more insight into, as well as empirical evidence of, the comparison of both approaches—machine learning and rule-based—in such deep IE tasks. As we compare both approaches, it will be important to pay attention to a number of different—and potentially conflicting—aspects: performance, development effort, scalability, computational efficiency, and so forth. A hybrid approach also could be explored. It may prove to be a good way of leveraging the best of both worlds.

## Conclusions and Contributions

This paper proposes a semantic deep IE method to automatically extract building energy requirements from contract specifications. A rule-based approach was used to develop the IE method, in which both syntactic features (POS tags, gazetteer features, and domain-specific tags) and semantic features (ontology concepts) are used to develop the extraction rules. To deal with the text complexities of specifications—hierarchically complex text structures, incomplete sentence structures, and the variety of LODs—three submethods are proposed: a domain-specific text splitting and stitching method, an incompleteness-aware sequential dependency extraction method, and a detail-aware LOD extraction method.

The proposed method was tested in extracting building thermal-insulation and lighting-power requirements from the MasterFormat specifications of an educational building project. A performance of 96.8% recall and 97.6% precision was achieved on the testing data. The experimental results indicate a number of conclusions. First, the proposed semantic IE method is promising in extracting energy requirements from specifications. Extraction errors may arise from dependency information, uncommon patterns, conflict resolution errors, extraction tool errors, ambiguous concept representation in the text, and/or coreference ambiguity. Second, text with hierarchically complex text structures successfully can be split and simplified in an automated way. Third, the use of incompleteness features, along with dependency information, is effective in reducing ambiguities. Fourth, the requirements related to a target LOD successfully can be filtered and extracted. A few errors occurred in tagging the action verbs of imperative sentences, which indicate LOD 400 or above, due to word-class ambiguity and incorrect exclusion verbs. However, these tagging errors did not propagate into IE errors; no IE errors were caused by the detail-aware LOD extraction.

The proposed method has potential good generalizability for three reasons. First, the proposed method could be generalized to other formats of specifications with necessary adaptions, because of the common text characteristics shared by different types of specifications. For example, incomplete sentence structures and streamlined writing style are shared by both MasterFormat and UniFormat. Second, the extraction rules developed are not specific to a particular concept that appeared only in the development text (e.g., "Foil-Faced_Rigid_Glass-Fiber_Board"). Instead, the most high-level concept for that particular concept (e.g., "Building_Envelope_Energy_Efficiency_Concept") was captured by the ontology and used in defining the syntactic and semantic feature patterns to improve the generalizability of rules. Therefore, the extraction rules can reason through the ontological concepts to identify the target information. Similarly, the extraction rules also are not specific to a particular pattern that appeared only in the development text; the patterns are generalized based on domain knowledge. Third, the extraction rules were developed based on five contract specifications, and tested on another contract's specifications to ensure that the rules are not overfitting to particular specifications. The promising performance of the experimental results indicates good generalizability of the proposed methods.

This work contributes to the body of knowledge in four primary ways. First, a semantic deep IE method is proposed to automatically extract energy requirements from contract specifications, thereby addressing a knowledge gap in the area of automated deep IE from specifications. In contrast to shallow methods, deep extraction can extract all information that describes a requirement, which is essential for ACC. Compared with code checking, which helps meet minimum regulatory requirements, specification checking is necessary to meet the additional requirements of the owner. Second, a domain-specific text splitting and stitching method is proposed to deal with hierarchically complex text structures. The proposed method uses a regular expressions-based pattern-matching technique to recognize the splitting signals automatically and split the text based on the text feature patterns. This allows for simplifying the hierarchically complex text structures and reducing the complexity of the text for further IE. Third, an incompleteness-aware sequential dependency extraction method is proposed to deal with incomplete sentence structures. The proposed method exploits incompleteness features, in addition to dependency information among SIEs, to reduce the text ambiguities. Fourth, a detail-aware LOD extraction method is proposed to deal with the variety of LODs. The LODs of the information are distinguished based on the grammatical moods of sentences. A domain-specific POS tagger is offered to support the recognition of domain-specific text features for use in analyzing the grammatical moods. After necessary adaptations, the proposed methods potentially are generalizable to other formats of specifications, as well as to other types of documents (e.g., OSHA standards) that share similar text characteristics and complexities (e.g., hierarchically complex text structures). The knowledge generated (e.g., incompleteness and LOD features) also could be integrated with machine learning approaches for a variety of document analytics tasks (e.g., extracting design parameters in different LODs for automated cost estimation), which would broaden the contribution of this effort to multiple areas of application within the civil engineering domain.

## Data Availability Statement

Some or all data, models, or code generated or used during the study are available from the corresponding author by request (extraction rules, tagging rules for domain-specific tags and for recognizing the action verbs, domain-specific POS tagger, and testing data set).


## Acknowledgments

The authors thank the National Science Foundation (NSF). This material is based upon work supported by the NSF under Grant No. 1201170. Any opinions, findings, and conclusions or recommendations expressed in this material are those of the authors and do not necessarily reflect the views of NSF.


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